

Hafeez Khan

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EDUCATION

- Florida Institute of Technology** Florida, USA
Masters & Ph.D. in Computer Science (PhD Advisor: Prof. Siddhartha Bhattacharyya) 2023 - 2027 (Expected)
- Birla Institute of Technology and Science, Pilani** Dubai, UAE
Bachelors of Engineering in Computer Science (Thesis Advisor: Prof. Raja Muthalaghu) 2019 - 2023

RECENT ACHIEVEMENT

1st place Winner, **Meta**-hosted Competition on Computationally Optimal Gaussian Splatting at ICCV 2025

Developed a lossy compression algorithm leveraging pruning and quantization, reducing 3DGS model sizes by **95.6%**, with near-optimal rendering performance on the FlashGS renderer.

RESEARCH EXPERIENCE

Incoming Systems Engineering Intern at Collins Aerospace Summer 2025
Guide: Janet Liu, Cong Liu **Collins Aerospace, USA**

- Developing an Agentic Vision-Language Model for runtime monitoring of aircraft in X-Plane simulation environment.

Research Assistant at ASSIST Lab (Florida Tech) Fall 2023 – Present
NASA-Funded Research Fall 2024 – Present
Guide: Natasha Neogi, Sarah Lehman, Siddhartha Bhattacharyya, Philip Chan **NASA LaRC, USA**

- Topics: Self-Supervised Learning, Multi-Modal Learning (Text and Vision, Video, Speech), Continual Learning
- Developed a Multi-Modal Self-Supervised Learning Model that integrates within the generative framework, and outperforms existing state-of-the-art on ImageNet-1k by **2.1%** Top-1 Accuracy.
- Created a continual learning model for lane detection, reaching optimal accuracy using just **41%** of parameters.
- Developed automated labeling algorithms to successfully label over **500,000** taxiways and runways images, improving detection accuracy by **7.31%** and reducing per-frame inference latency by **58.4%**.

Microsoft-Collaboration Research Spring 2025
Guide: Yash Jain, Vibhav Vineet **Microsoft Research, USA**

- Topics: Image Generation, Multi-modal Large Language Models, Text-to-Image, Prompt Optimization
- Developed a closed-loop test-time prompt refinement framework using multi-modal LLMs, improving text-to-image alignment by **11.7%** on DALL-E 3, **12.9%** on Flux, and upto **15.25%** on Stable Diffusion 1.5/2.1/3.0 models.

DARPA-Funded Research Summer 2025
Guide: Junaid Babar, Isaac Amundson, Siddhartha Bhattacharyya **Collins Aerospace, USA**

- Topics: Trust and Safety, Formal Verification, Reinforcement Learning, Model Checking, Language Translation
- Built an automated Soar-to-PRISM translator in ANTLR/Java to enable formal verification of cognitive models.

American Heart Association-Funded Research Summer 2024
Guide: Venkat Keshav Chivrukula **UTHealth Houston, USA**

- Topics: 3D/4D Reconstruction, Representation Learning, Computational Fluid Dynamics (CFD)
- Developed an autoencoder for 3D intraventricular velocity reconstruction, surpassing UNet3D by **10.55%** in PSNR.
- Generated synthetic CFD hemodynamic datasets across **8** LVAD patient geometries using ANSYS Fluent for training 3D models.

AWARDS

- Awarded for **Outstanding Academic Achievement** at Florida Tech. 2025
- Undergraduate Research Award** for outstanding Bachelors Thesis at BITS Pilani Dubai. 2023
- Gold Medalist** in IEEE Robotics Competition, 3D Object Detection and Tracking for Road Safety, held in Dubai. 2023

SKILLS

Programming Languages: Python, R, Matlab, C/C++, Java, SQL

Proficient Deep Learning and Vision Libraries: PyTorch, TensorFlow, Diffusers, MMDetection, OpenCV, Open3D

Cloud Platforms & Services: Azure (ML Studio, DevOps), AWS (S3, SageMaker), GCP (Google Maps, Cloud Storage).

PUBLICATIONS

* equal contribution

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift [\[PDF\]](#)

[M.A. Hafeez Khan](#), Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
[\[Oral Presentation\]](#) International Conference on Computer Vision 2COOL Workshop (**ICCV**), 2025

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding in Diverse Conditions [\[PDF\]](#)

Parth Ganeriwala, [M.A. Hafeez Khan](#), Amy Alvarez, Siddhartha Bhattacharyya, Natasha Neogi, Sarah M. Lehman
American Institute of Aeronautics and Astronautics (**AIAA**) SCITECH Forum, 2026

Test-time Prompt Refinement for Text-to-Image Models [\[PDF\]](#)

[M.A. Hafeez Khan*](#), Yash Jain*, Siddhartha Bhattacharyya, Vibhav Vineet
[\[Invited Talk\]](#) International Conference on Computer Vision MARS2 Workshop (**ICCV**), 2025

Few-Shot Classification and Anatomical Localization of Tissues in SPECT Imaging [\[PDF\]](#)

[M.A. Hafeez Khan](#), Samuel M. Boddepalli, Siddhartha Bhattacharyya, Debasis Mitra
[\[Oral Presentation\]](#) Medical Imaging Conference (**MIC**), 2025

LVADNet3D: A Deep Autoencoder for Reconstructing 3D Intraventricular Flow from Sparse Data [\[PDF\]](#)

[M.A. Hafeez Khan](#), Marcello Mattei, Ben Diaz, Ruth White, Siddhartha Bhattacharyya, Venkat Keshav Chivukula
[\[Oral Presentation\]](#) International Conference on Machine Learning and Applications (**ICMLA**), 2025

NORA: A Nephrology-Oriented Representation Learning Approach Towards CKD Classification [\[PDF\]](#)

[M.A. Hafeez Khan](#), T. Bhattacharyya, O. Khan, Noorah Khan, Alina Khan, M.Q. Khan, Sujoy Hajra
[\[Spotlight Paper\]](#) International Conference on Machine Learning and Applications (**ICMLA**), 2025

ALINA: Advanced Line Identification and Notation Algorithm [\[PDF\]](#)

[M.A. Hafeez Khan](#), Parth Ganeriwala, Siddhartha Bhattacharyya, Natasha Neogi, Raja Muthalagu
IEEE / CVF Computer Vision and Pattern Recognition Conference (**CVPR**), 2024

A Hybrid BiLSTM-CNN Approach for Intrusion Detection for IoT Applications [\[PDF\]](#)

[M.A. Hafeez Khan](#), Sapna Sadhwani, Raja Muthalagu, Pranav Pawar, K. Suresh
Scientific Reports, Springer, 2024

AssistTaxi: A Comprehensive Dataset for Taxiway Analysis and Autonomous Operations [\[PDF\]](#)

Parth Ganeriwala, Siddhartha Bhattacharyya, S. Gunther, Brian Kish, [M.A. Hafeez Khan](#), Aknur Dhadoti, Natasha Neogi
International Conference on Machine Learning and Applications (**ICMLA**), 2023

Classification of Microstructure Images of Metals Using Transfer Learning [\[PDF\]](#)

[M.A. Hafeez Khan](#), Hrishikesh Sabnis, J. Angel Arul Jothi, J. Kanishkha, A.D. Prasad
International Conference on Modelling and Development of Intelligent Systems (**MDIS**), 2022

Detection of Cavities from Oral Images using Convolutional Neural Networks [\[PDF\]](#)

[M.A. Hafeez Khan](#), Giri Prasad S., J. Angel Arul Jothi
[\[Best Paper Award\]](#) IEEE International Conference on Electrical, Computer and Energy Technologies (**ICECET**), 2022

Detection of Bicep Form Using Myoware and Machine Learning [\[PDF\]](#)

[M.A. Hafeez Khan](#), Rohan V. Rudraraju, R. Swarnalatha
International Conference on Advances in Data-driven Computing and Intelligent Systems (**ADCIS**), 2022

TEACHING EXPERIENCE

Graduate Teaching Assistant

Florida Tech

Primary Instructor: Prof. Philip Chan (FLTech)

Aug '23 – May '24

- Course: CSE2010 Algorithms & Data Structures; My Rating: 4.9/5.